

Week 3: Lecture and In-Class Problems

Linear Algebra I · Fall 2026

Matrices, determinants, and scalar triple products

80-minute lecture + 90-minute problem session.

Part I develops the theory with explicit definitions and worked examples. Part II contains 15 problems, increasing from basic calculations to challenges.

Suggested lecture timing: recap 5 min; entrywise operations and transpose 15 min; matrix multiplication 20 min; determinants 20 min; triple product and volume 15 min; questions 5 min.

For the 90-minute practice session: problems 1–4 (15 min), 5–8 (25 min), 9–12 (30 min), then selected challenges 13–15 (20 min). Use uncompleted challenges for follow-up discussion.

All vectors are displayed as columns. Each problem begins on a new PDF page. A wide blank space separates each statement from its solution. Figures are generated from the stated coordinates; the notebook contains the executable Python and embedded plots.

Week 2 recap — one page

Vectors are columns. For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the dot product and Euclidean length are

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \sum_{i=1}^n u_i v_i, \quad \|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}.$$

For nonzero vectors, $\cos \theta = (\mathbf{u} \cdot \mathbf{v})/(\|\mathbf{u}\|\|\mathbf{v}\|)$. A zero dot product means perpendicularity; angles involving the zero vector are undefined.

In $\mathbb{R}^3$, the cross product is a vector perpendicular to its two inputs:

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}.$$

Its magnitude is parallelogram area; reversing the order reverses the sign. For example,

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \quad \mathbf{u} \times \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix}.$$

An $m \times n$ matrix maps a column in $\mathbb{R}^n$ to a column in $\mathbb{R}^m$:

$$A\mathbf{x} = x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n.$$

Here $\mathbf{a}_j$ is column j of A . This column-combination picture will explain matrix multiplication and determinant geometry this week.

1. Matrix size, addition, and scalar multiplication

A matrix is a rectangular array. The first dimension counts rows and the second counts columns; a_{ij} is the entry in row i , column j . Always check sizes before calculating.

Addition and subtraction require the **same size**. They act entry by entry, as does multiplication by a scalar:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = ca_{ij}.$$

For the two 2×3 matrices

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \end{bmatrix},$$

we obtain

$$2A - B = \begin{bmatrix} 0 & -4 & -1 \\ 7 & -1 & 6 \end{bmatrix}.$$

The $(2, 3)$ entry, for example, is $2(4) - 2 = 6$.

The zero matrix has all entries zero. The square identity I_n has ones on its diagonal and zeros elsewhere; it leaves every column vector unchanged. A diagonal matrix scales coordinates separately. A triangular matrix has zeros entirely above or entirely below its diagonal.

Ask the class: can a 2×3 matrix be added to a 3×2 matrix? No: matching total numbers of entries are insufficient.

2. Transpose: rows become columns

The transpose interchanges the two indices: $(A^T)_{ij} = a_{ji}$. Thus an $m \times n$ matrix becomes an $n \times m$ matrix. It does not change the entries' signs.

$$A = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 3 & 5 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 0 \\ -2 & 3 \\ 4 & 5 \end{bmatrix}.$$

Transposing twice returns the original matrix. Addition and scalar multiplication commute with transposition:

$$(A^T)^T = A, \quad (A + B)^T = A^T + B^T, \quad (cA)^T = cA^T.$$

A square matrix is **symmetric** when $A^T = A$. Its entries mirror each other across the diagonal, for example

$$H = \begin{bmatrix} 2 & -1 & 3 \\ -1 & 4 & 0 \\ 3 & 0 & 5 \end{bmatrix}.$$

Our vectors remain columns. The notation $\mathbf{u}^T$ explicitly denotes the row obtained by transposing a column. This is why $\mathbf{u}^T \mathbf{v}$ can produce a scalar.

Check: the diagonal is unchanged by transposition. A nonsquare matrix cannot equal its transpose, because their sizes differ.

3. Matrix multiplication: dimensions and dot products

Matrix multiplication represents one linear operation following another. It is different from multiplying matching entries.

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{n \times p} \quad \implies \quad AB \in \mathbb{R}^{m \times p}.$$

The inner dimensions must match. Each output entry is one row-column dot product:

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

For example,

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & -1 \\ 3 & 2 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1 & -4 \\ 9 & 1 \end{bmatrix}.$$

The first entry is $1(2) + 2(1) - 1(3) = 1$, and the bottom-right entry is $0(0) + 3(-1) + 2(2) = 1$.

Here AB is 2×2 , whereas BA is 3×3 . Both products exist, but they cannot be equal because their sizes differ. In other examples, only one order may be defined.

4. The column picture of a product

Write B as a collection of columns. Then A acts on each column separately:

$$B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_p], \quad AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_p].$$

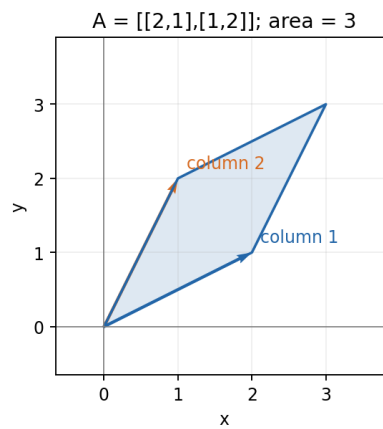
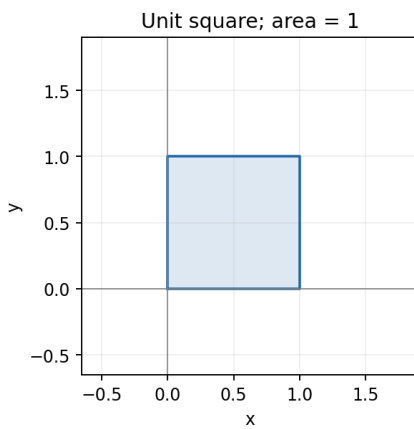
Each column of AB is a combination of the columns of A , using the entries of the corresponding column of B as weights.

For example, let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Then $A\mathbf{x} = \mathbf{a}_1 + 2\mathbf{a}_2 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$. The images of the standard basis columns are exactly the columns of A .

Consequently, the unit square becomes the parallelogram shown below. Its area is 3; we will recover this using a determinant.



5. Composition: the rightmost matrix acts first

For a column $\mathbf{x}$, the product $DS\mathbf{x}$ means $D(S\mathbf{x})$. Apply S first, then D . Parentheses can move without changing the order: $(AB)C = A(BC)$.

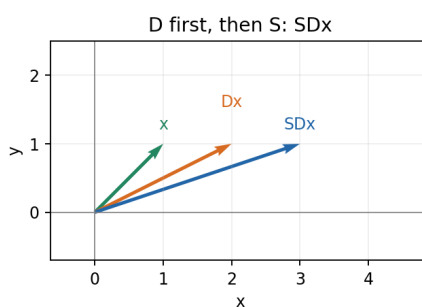
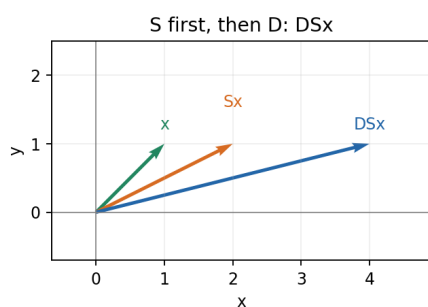
Consider a shear and a horizontal stretch:

$$S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The two orders give

$$DS = \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix}, \quad SD = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \quad DS\mathbf{x} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad SD\mathbf{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Thus matrix multiplication is generally **not commutative**. Both orders below use the same input and the same two transformations.



6. Product rules and transposing a product

For compatible sizes, multiplication is associative and distributes over addition:

$$A(BC) = (AB)C, \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC.$$

For A of size $m \times n$, the identity rule is $I_m A = A = A I_n$. Keep factors in their original order when expanding expressions.

Transposing a product **reverses the order**:

$$(AB)^T = B^T A^T.$$

To see why, its (i, j) entry is $(AB)_{ji} = \sum_k a_{jk} b_{ki}$. The (i, j) entry of $B^T A^T$ is the same sum, $\sum_k b_{ki} a_{jk}$.

For two columns in $\mathbb{R}^2$,

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ -1 \end{bmatrix},$$

the **inner product** is the scalar $\mathbf{u}^T \mathbf{v} = 1$. The **outer product** is the matrix

$$\mathbf{u} \mathbf{v}^T = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}.$$

Sizes distinguish these operations: $(1 \times 2)(2 \times 1)$ versus $(2 \times 1)(1 \times 2)$. Also $A^T A$ is symmetric, since $(A^T A)^T = A^T A$.

7. A 2×2 determinant measures signed area

A determinant is a scalar associated with a **square** matrix. For two dimensions,

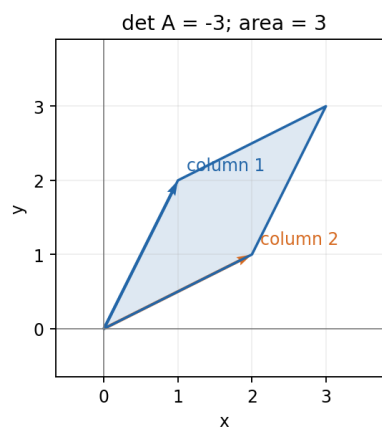
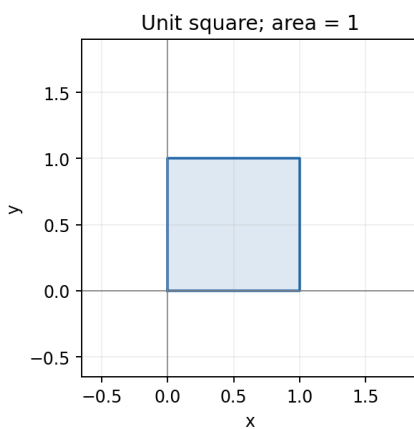
$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det A = ad - bc.$$

The columns are the two edges of the transformed unit square. Its area is $|\det A|$. A positive determinant preserves the usual orientation; a negative one reverses it. A zero determinant collapses the square to a line or point.

For the plotted example,

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad \det A = 1 - 4 = -3.$$

The area is 3 square units, not -3 . The negative sign records the reversed order of the two column directions. The triangle with vertices $\mathbf{0}, \mathbf{a}_1, \mathbf{a}_2$ has half the parallelogram area.



8. Cofactors: a systematic 3×3 calculation

Let M_{ij} be the submatrix left after deleting row i and column j . Its determinant is the corresponding minor. The cofactor is $C_{ij} = (-1)^{i+j} \det M_{ij}$. The sign pattern is

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Expand along any one row or column. For the first row,

$$\det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}.$$

Choose a row or column with zeros to reduce arithmetic. For

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix},$$

the first-row expansion is

$$\det A = 2 \det \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} - \det \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} + 0 = 12 - (-1) = 13.$$

The entry $a_{12} = 1$ is positive, but its cofactor carries a minus sign. Keep these two signs separate.

For any upper or lower triangular matrix, the determinant is the product of the diagonal entries. A diagonal example is $\det \text{diag}(2, -3, 4) = -24$. Diagonal multiplication alone is not valid for a general matrix.

9. Determinant rules: simplify before expanding

The following rules hold for square matrices; the row rules also hold for columns.

Change or operation	Effect on the determinant
Exchange two rows	Reverse its sign
Multiply one row by c	Multiply by c
Add a multiple of one row to another	No change
A zero row or two proportional rows	Determinant is zero
Transpose	$\det(A^T) = \det A$
Product of equally sized square matrices	$\det(AB) = \det A \det B$

Scaling an entire $n \times n$ matrix scales all n rows:

$$\det(cA) = c^n \det A.$$

For instance, if a 3×3 matrix has determinant 5, doubling every entry gives determinant $2^3(5) = 40$.

Determinants do **not** distribute over matrix addition. For $A = B = I_2$, the determinant of $A+B = 2I_2$ is 4, whereas $\det A + \det B = 2$.

Geometrically, exchanging two edges reverses orientation. Adding a multiple of one edge to another shears a parallelogram or parallelepiped without changing its area or volume.

10. Scalar triple product: definition and calculation

Define three columns in three-dimensional Euclidean space:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}.$$

First take a cross product, then a dot product. The output is a **scalar**:

$$T = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}].$$

The brackets on the right contain the vectors as columns. A cyclic permutation leaves T unchanged; exchanging two inputs reverses its sign. Do not confuse this with a vector triple product such as $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$.

For the numerical example used on the next page, take

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Then $\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix}$ and $T = 12$. Equivalently, the matrix with these columns is upper triangular, with determinant $2(3)(2) = 12$.

11. Volume, height, and orientation in xyz space

For the three edges

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix},$$

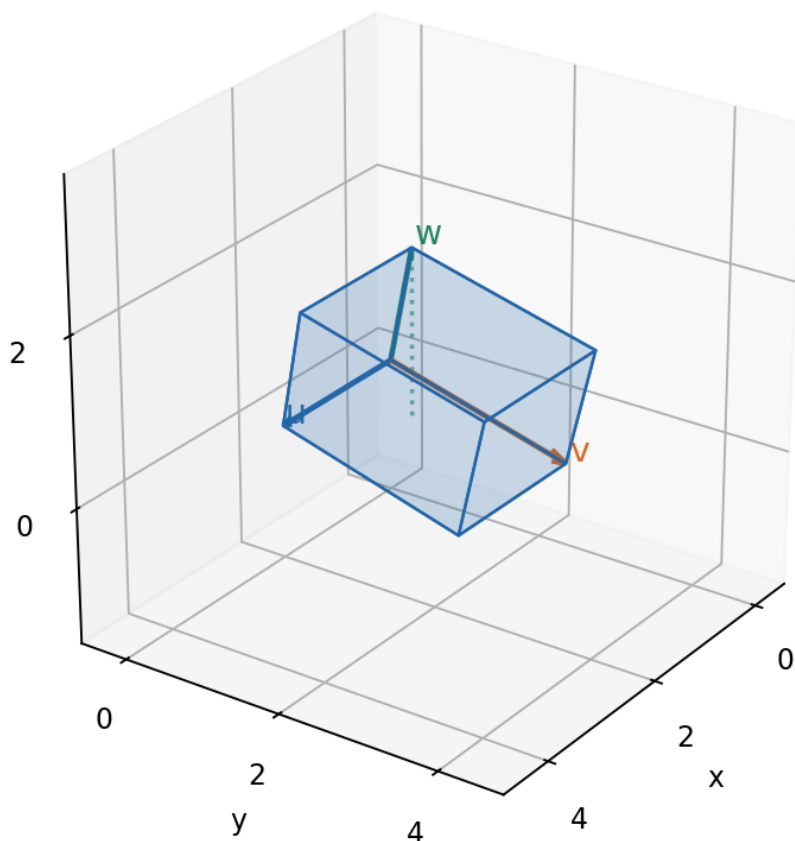
the parallelepiped consists of all $s\mathbf{u} + t\mathbf{v} + r\mathbf{w}$ with $0 \leq s, t, r \leq 1$. Its volume is $|T| = 12$ cubic units.

Here the base area is $\|\mathbf{u} \times \mathbf{v}\| = 6$ and perpendicular height is 2. In general, if the base is nondegenerate,

$$h = \frac{|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|}{\|\mathbf{u} \times \mathbf{v}\|}, \quad V = (\text{base area})h = |T|.$$

The tilted third edge need not be perpendicular to the base. The plot uses the actual coordinates with equal scale on the three axes; dotted guides locate edge tips relative to the xy plane.

Base area 6; height 2; volume 12



12. Coplanarity and tetrahedra

Three vectors are **coplanar** when they lie in some plane through the origin. Their scalar triple product is then zero: there is no three-dimensional volume. Conversely, $T = 0$ means the three columns lie in a plane (or a line).

For points P, Q, R, S , use edges from the same starting point:

$$\mathbf{u} = Q - P, \quad \mathbf{v} = R - P, \quad \mathbf{w} = S - P.$$

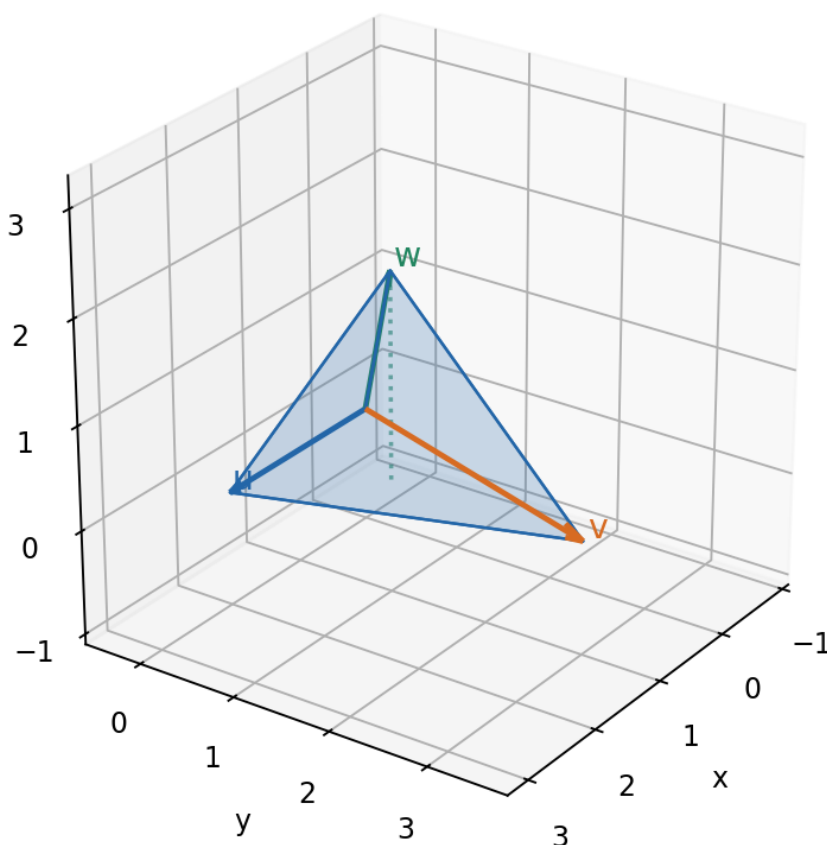
The tetrahedron with those four vertices has volume

$$V_{\text{tetrahedron}} = \frac{1}{6} |\det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]|.$$

This is one sixth of the corresponding parallelepiped. The base triangle has half the parallelogram area, and a pyramid has one third of base area times height.

In the figure, $P = \mathbf{0}$, $Q = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$, $R = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$, and $S = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$. Its volume is $12/6 = 2$.

Tetrahedron: volume 2



Reading and end-of-theory checklist

Reading: Gilbert Strang, *Introduction to Linear Algebra*, 4th edition: §2.4 (matrix operations), §2.7 (transpose), §§5.1–5.2 (determinant properties and cofactors), and the volume, cross-product, and scalar-triple-product portions of §5.3. These topics follow Week 3 of the updated course syllabus. Inverses and rank are reserved for Week 4.

Before starting the problems, be able to explain:

1. Why the dimensions of a product determine whether it exists.
2. Why $AB\mathbf{x}$ applies B before A , and why order matters.
3. Why $(AB)^T = B^T A^T$.
4. The difference between a determinant and an area or volume.
5. How to expand a determinant while keeping cofactor signs correct.
6. Why $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ detects coplanarity.

Common checks: show every vector as a column; label matrix sizes; retain product order; take an absolute value for area or volume; form edges from one common vertex for a tetrahedron.

Part II: In-class problems

The theory section ends here.

Take a short pause before beginning the problem session. Work through the basic problems first, then the applications, and finally selected challenges.

Try each problem before reading the solution. Use the geometric figures to check the algebra.

Sequence: matrix operations (1–8), determinants (9–11), volume (12), and challenges (13–15).

Problem 1: Entrywise operations and dimensions — basic

Let $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$. Compute $2A - B$ and identify its $(2, 1)$ entry. Can A be added to a 2×3 matrix? Explain.

Solution 1

Both matrices are 2×2 , so subtract corresponding entries:

$$2A - B = \begin{bmatrix} 0 & -5 \\ 7 & -4 \end{bmatrix}.$$

The $(2, 1)$ entry is $2(3) - (-1) = 7$. Addition to a 2×3 matrix is undefined: both dimensions must match.

Problem 2: Transpose and symmetric products — basic

Let $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$. Write C^T and state its size. Compute CC^T and decide whether it is symmetric.

Solution 2

The transpose has size 3×2 :

$$C^T = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ -1 & 2 \end{bmatrix}, \quad CC^T = \begin{bmatrix} 6 & 4 \\ 4 & 13 \end{bmatrix}.$$

Each diagonal entry is a row dotted with itself; each off-diagonal entry is one row dotted with the other. The result is symmetric. More generally, $(CC^T)^T = (C^T)^T C^T = CC^T$.

Problem 3: Matrix times a column — basic

For $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, compute $C\mathbf{x}$ using both row dot products and a combination of columns.

Solution 3

The sizes are $(2 \times 3)(3 \times 1)$, so the answer is a 2×1 column:

$$C\mathbf{x} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

The column calculation gives the same result:

$$(2) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 3 \end{bmatrix} + (1) \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

The two output entries are the dot products of the two rows of C with $\mathbf{x}$.

Problem 4: Does the order matter? — basic

Let $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$. Compute AB and BA . Are they equal?

Solution 4

Use a row from the first factor and a column from the second:

$$AB = \begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}, \quad BA = \begin{bmatrix} 5 & -4 \\ 11 & 2 \end{bmatrix}.$$

For example, $(AB)_{11} = (1)(2) + (-2)(-1) = 4$. The products differ. Even when both orders have the same size, equality is not guaranteed.

Problem 5: Two rectangular products — intermediate

Let $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ 1 & -1 \\ 3 & 2 \end{bmatrix}$. Predict the sizes of CD and DC , then compute both.

Solution 5

The inner dimensions match in each order. The outer dimensions give CD of size 2×2 and DC of size 3×3 :

$$CD = \begin{bmatrix} 1 & -4 \\ 9 & 1 \end{bmatrix}, \quad DC = \begin{bmatrix} 2 & 4 & -2 \\ 1 & -1 & -3 \\ 3 & 12 & 1 \end{bmatrix}.$$

Each column of CD can also be computed by applying C to the corresponding column of D . The two products cannot be compared as equal matrices because their sizes differ.

Problem 6: Composition of geometric transformations — intermediate

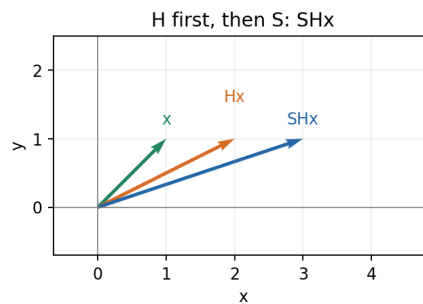
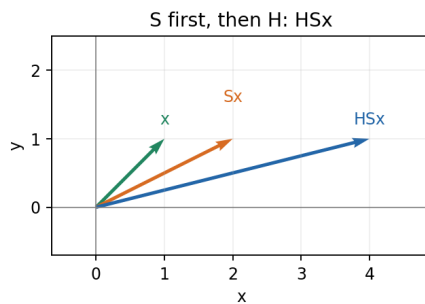
Let $S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $H = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$, and $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Compute $HS\mathbf{x}$ and $SH\mathbf{x}$. Describe the transformation performed by each matrix and explain the difference.

Solution 6

S sends $\begin{bmatrix} x \\ y \end{bmatrix}$ to $\begin{bmatrix} x + 1y \\ y \end{bmatrix}$, a horizontal shear. H multiplies the horizontal coordinate by 2.

$$S\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad HS\mathbf{x} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad H\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad SH\mathbf{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

In $HS\mathbf{x}$, the shear contribution is stretched as well; in $SH\mathbf{x}$, the shear is added after stretching.



Problem 7: Inner and outer products — intermediate

For $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$, compute $\mathbf{u}^T \mathbf{v}$ and $\mathbf{u} \mathbf{v}^T$. State the size of each result and verify $(\mathbf{u} \mathbf{v}^T)^T = \mathbf{v} \mathbf{u}^T$.

Solution 7

The inner product is a scalar (a 1×1 matrix):

$$\mathbf{u}^T \mathbf{v} = (1)(2) + (2)(0) + (-1)(3) = -1.$$

The outer product is 3×3 :

$$\mathbf{u} \mathbf{v}^T = \begin{bmatrix} 2 & 0 & 3 \\ 4 & 0 & 6 \\ -2 & 0 & -3 \end{bmatrix}, \quad \mathbf{v} \mathbf{u}^T = \begin{bmatrix} 2 & 4 & -2 \\ 0 & 0 & 0 \\ 3 & 6 & -3 \end{bmatrix}.$$

The displayed matrices are transposes of one another. Their entries are $u_i v_j$ and $v_i u_j$, respectively.

Problem 8: Verify the reverse-order transpose rule — intermediate

Use $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$. Compute $(AB)^T$ and $B^T A^T$ independently. Does $A^T B^T$ give the same answer?

Solution 8

Multiplying first and then transposing gives

$$AB = \begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}, \quad (AB)^T = \begin{bmatrix} 4 & 6 \\ -7 & 3 \end{bmatrix}.$$

In reverse order,

$$B^T A^T = \begin{bmatrix} 4 & 6 \\ -7 & 3 \end{bmatrix}.$$

But $A^T B^T = \begin{bmatrix} 5 & 11 \\ -4 & 2 \end{bmatrix}$, which is different. The correct identity reverses the factor order.

Problem 9: Determinant, area, and orientation — intermediate

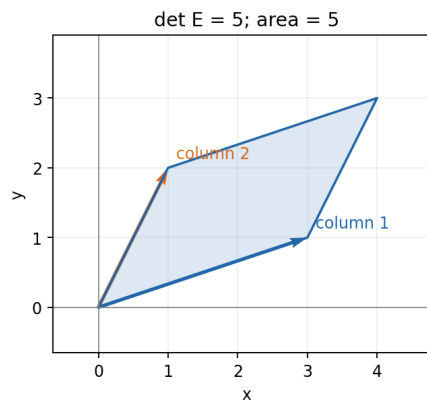
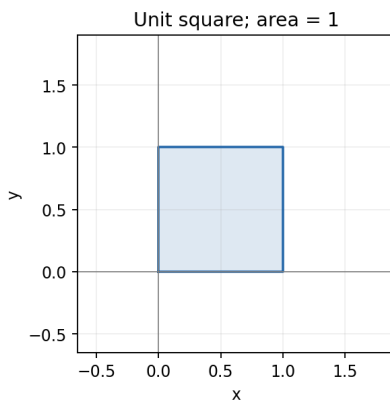
Let $E = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$. Compute its determinant. Find the area of the parallelogram formed by its columns and of the triangle with vertices $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2$, where $\mathbf{e}_1, \mathbf{e}_2$ denote those columns. Does E preserve orientation?

Solution 9

The determinant is

$$\det E = (3)(2) - (1)(1) = 5.$$

The parallelogram area is 5 square units, and the triangle area is $5/2$. The determinant is positive, so orientation is preserved. Area is always nonnegative.



Problem 10: A 3×3 determinant by cofactors — intermediate

Compute $\det F$ for $F = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ by expanding along the first row. Then check your answer by expanding along the first column.

Solution 10

The first-row signs are $+, -, +$. Since the last entry is zero,

$$\det F = (2)(6) - (1)(-1) = 13.$$

The first-column expansion gives

$$(2) \det \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} + (1) \det \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = 13.$$

The middle term vanishes. The $(3, 1)$ cofactor has a positive sign, since $(-1)^{3+1} = 1$.

Problem 11: Determinant rules without expansion — intermediate

A 3×3 matrix M satisfies $\det M = 5$. Starting from M , form N by swapping rows 1 and 2, then multiplying row 3 by 2, then adding 3 times the current row 1 to row 2. Find $\det N$, $\det(2M)$, and $\det(M^T M)$.

Solution 11

The row swap changes the sign, row scaling multiplies by 2, and the final row addition leaves the determinant unchanged:

$$\det N = -2(5) = -10.$$

Scaling all three rows gives $\det(2M) = 2^3 \det M = 40$. Finally,

$$\det(M^T M) = \det M^T \det M = (\det M)^2 = 25.$$

Distinguish scaling one row from scaling the whole matrix.

Problem 12: Scalar triple product and volume — intermediate

Let $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, and $\mathbf{w} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$. Compute $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$, the parallelepiped volume, and the tetrahedron volume for vertices $\mathbf{0}, \mathbf{u}, \mathbf{v}, \mathbf{w}$. What changes if $\mathbf{u}$ and $\mathbf{v}$ are exchanged?

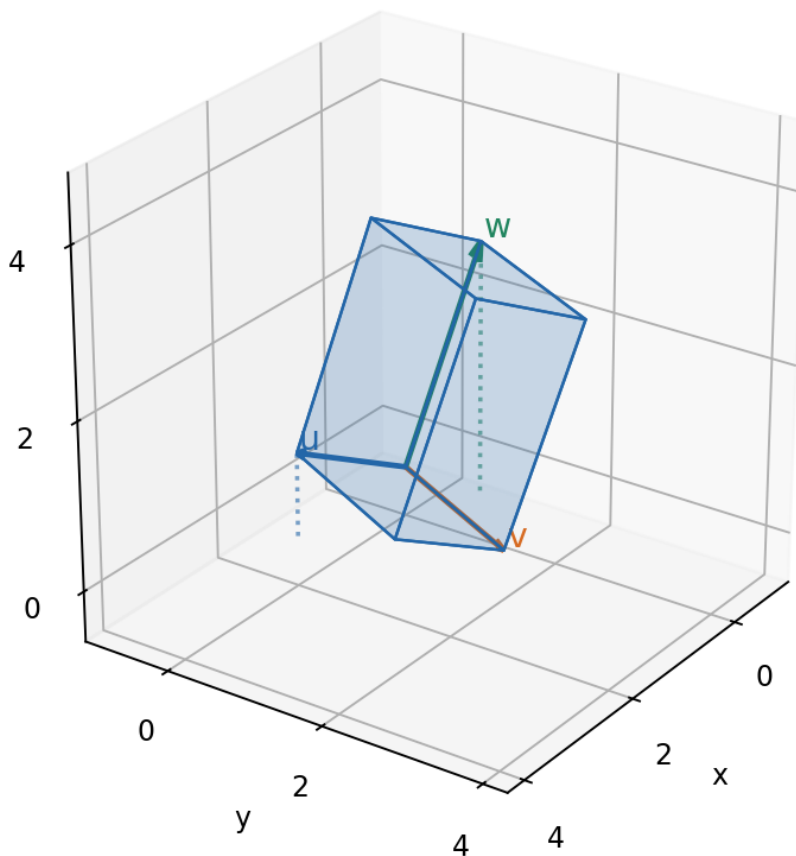
Solution 12

First form the cross product and then the dot product:

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}, \quad (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = 13.$$

Thus the parallelepiped volume is 13 and the tetrahedron volume is $13/6$ cubic units. Exchanging the first two inputs changes the scalar triple product to -13 , but both geometric volumes stay the same.

Parallelepiped; volume = 13



Problem 13: Find a parameter for coplanarity — challenge

For $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$, and $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ k \end{bmatrix}$, find all real k for which the vectors are coplanar. At that value, express $\mathbf{w}$ as a combination of $\mathbf{u}$ and $\mathbf{v}$.

Solution 13

The cross product is $\mathbf{u} \times \mathbf{v} = \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$. Therefore

$$T = -2 - 1 + 2k.$$

Coplanarity requires $T = 0$, so $k = \frac{3}{2}$. Comparing the first two coordinates fixes the coefficients:

$$\mathbf{w} = \mathbf{u} + \frac{1}{2}\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 3/2 \end{bmatrix}.$$

The third coordinate agrees exactly at the stated value of k .

Problem 14: Which matrices commute with a diagonal matrix? — challenge

Let $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ and $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Find every X such that $DX = XD$. What changes if both diagonal entries of D are equal?

Solution 14

Direct multiplication gives

$$DX = \begin{bmatrix} 2a & 2b \\ 3c & 3d \end{bmatrix}, \quad XD = \begin{bmatrix} 2a & 3b \\ 2c & 3d \end{bmatrix}.$$

Equality forces $1b = 0$ and $1c = 0$. Hence $b = c = 0$, while a, d are arbitrary: precisely the diagonal matrices commute with this D . If $D = \lambda I$, then $DX = \lambda X = XD$ for every X .

Problem 15: How a transformation changes volume — challenge

Let $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$, and $L = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$. Without computing three new edge vectors, find the signed triple product and volume after applying L to every edge. Explain why the volume factor is the same for every initial parallelepiped.

Solution 15

Put $U = [\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$. Its determinant is 13. The transformed column matrix is LU , so

$$\det(LU) = \det L \det U = (-2)(13) = -26.$$

The new volume is 26 cubic units. The factor $|\det L| = 2$ is independent of U , while the negative sign reverses orientation. This gives a geometric meaning to the product rule for determinants.